Amazon 데이터셋 기반 추천 시스템 설계 프로젝트 - 신정아

프로젝트 개요 : 이번 프로젝트는 Amazon 데이터셋을 기반으로 고객 맞춤형 추천 시스템을 설계하고 구현하는 작업이다. SQL만 사용하여 분석하고, 고객의 구매 경험을 향상시킬 수 있는 추천 시스템의 기초를 설계하는 것이 이번 프로젝트의 목표이다.

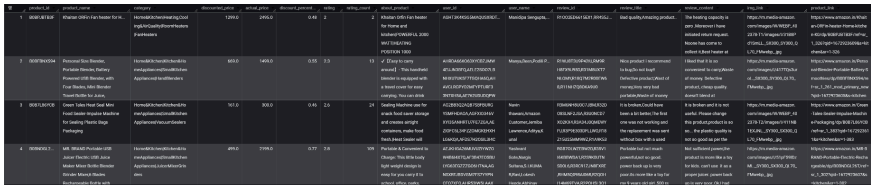
13-1. 데이터 불러오기

데이터 살펴보기

- **테이블에 있는 5개의 행만 출력하기**

```
-- 테이블 불러오기 : (Tip) 데이터세트에서 테이블명 우측 (:)을 클릭 후 '쿼리' 선택
SELECT *
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
LIMIT 5
```

[결과 이미지]



- **전체 데이터 구조 확인하기 (전체 행수 확인) : 1,465행**

```
SELECT COUNT(*) AS Total_Rows
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
```

[결과 이미지]

행	Total_Rows
1	1465

- **결측치 현황 : rating_count 컬럼 결측치 2개 확인**

```
SELECT
       SUM(CASE WHEN product_id IS NULL THEN 1 ELSE 0 END) AS product_id_null,
       SUM(CASE WHEN product_name IS NULL THEN 1 ELSE 0 END) AS product_name_null,
       SUM(CASE WHEN category IS NULL THEN 1 ELSE 0 END) AS category_null,
       SUM(CASE WHEN discounted_price IS NULL THEN 1 ELSE 0 END) AS discounted_price_null,
       SUM(CASE WHEN actual_price IS NULL THEN 1 ELSE 0 END) AS actual_price_null,
       SUM(CASE WHEN discount_percentage IS NULL THEN 1 ELSE 0 END) AS discount_percentage
_null,
       SUM(CASE WHEN rating IS NULL THEN 1 ELSE 0 END) AS rating_null,
       SUM(CASE WHEN rating_count IS NULL THEN 1 ELSE 0 END) AS rating_count_null,
       SUM(CASE WHEN about_product IS NULL THEN 1 ELSE 0 END) AS about_product_null,
       SUM(CASE WHEN user_id IS NULL THEN 1 ELSE 0 END) AS user_id_null,
       SUM(CASE WHEN user_name IS NULL THEN 1 ELSE 0 END) AS user_name_null,
       SUM(CASE WHEN review_id IS NULL THEN 1 ELSE 0 END) AS review_id_null,
       SUM(CASE WHEN review_title IS NULL THEN 1 ELSE 0 END) AS review_title_null,
       SUM(CASE WHEN review_content IS NULL THEN 1 ELSE 0 END) AS review_content_null,
       SUM(CASE WHEN img_link IS NULL THEN 1 ELSE 0 END) img_link_null,
```

```
        SUM(CASE WHEN product_link IS NULL THEN 1 ELSE 0 END) AS product_link_null
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
```

[결과 이미지]

행	product_id_n...	product_name...	category_null	discounted_price...	actual_price_null	discount_percent...	rating_null	rating_count_null	about_product_null	user_id_null	user_name_null	review_id_null	review_title_n...	review_content_n...	img_link_null	product_link_nu
1	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	

행	Total_Rows
1	1465

데이터 EDA

- **Amazon 플랫폼에서 판매되는 제품의 수(데이터셋 상) : 1,351건**

```
SELECT COUNT(DISTINCT product_id) AS product_count
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
```

[결과 이미지]

행	product_count
1	1351

- **Amazon 플랫폼에서 판매되는 제품의 수(product_name 기준) : 1,337건**

```
SELECT COUNT(DISTINCT product_name) AS product_count
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
```

[결과 이미지]

행	product_count
1	1337

※ 14건의 중복 케이스(동일 product_name ≠ product_id)

(추가 확인) 동일 product_name에 다른 product_id 존재 내역 확인

```
SELECT product_name, COUNT(DISTINCT product_id)
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
GROUP BY 1
ORDER BY COUNT(DISTINCT product_id) DESC;
```

[결과 이미지] 9개 상품명에 중복 product_id 존재 확인

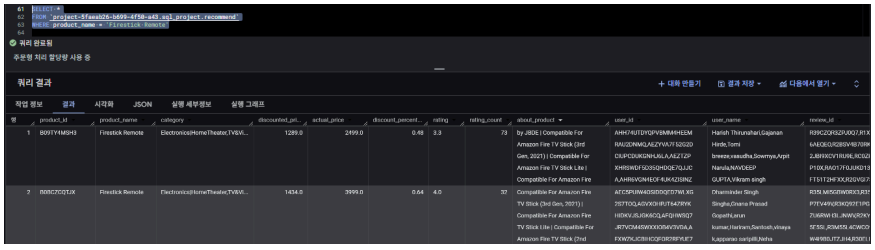
행	product_name	f0_
1	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Watch with Bluetooth Calling, AI Voice Assistance, 100 Sports Modes IP67 Rating, 240*280 Pixel High Resolution	5
2	Fire-Boltt Phoenix Smart Watch with Bluetooth Calling 1.3",120+ Sports Modes, 240*240 PX High Res with SpO2, Heart Rate Monitoring & IP67 Rating	3
3	Samsung Galaxy M13 5G (Aqua Green, 6GB, 128GB Storage) | 5000mAh Battery | Upto 12GB RAM with RAM Plus	3
4	Firestick Remote	2
5	Fire-Boltt Visionary 1.78" AMOLED Bluetooth Calling Smartwatch with 368*448 Pixel Resolution 100+ Sports Mode, TWS Connection, Voice Assistance, SPO2 & Heart Rate Monitoring	2
6	Smashtronics® - Case for Firetv Remote, Fire Stick Remote Cover Case, Silicone Cover for TV Firestick 4K/TV 2nd Gen(3rd Gen) Remote Control - Light Weight/Anti Slip/Shockproof (Black)	2
7	Fire-Boltt India's No 1 Smartwatch Brand Talk 2 Bluetooth Calling Smartwatch with Dual Button, Hands On Voice Assistance, 60 Sports Modes, In Built Mic & Speaker with IP68 Rating	2
8	Samsung Galaxy M33 5G (Emerald Brown, 6GB, 128GB Storage) | 6000mAh Battery | Upto 12GB RAM with RAM Plus | Travel Adapter to be Purchased Separately	2
9	Redmi Note 11 (Horizon Blue, 6GB RAM, 64GB Storage)|90Hz FHD+ AMOLED Display | Qualcomm® Snapdragon™ 680-6nm | 33W Charger Included	2
10	Tecno Spark 8T (Turquoise Cyan, 4GB RAM,64GB Storage) | 50MP AI Camera | 7GB Expandable RAM	1

※ 9개의 중복 product_name 세부 내역 확인

```
# 5개의 product_id 존재
# discounted_price, actual_price, discount_percentage, rating,rating_count 등
# link관련 부분을 제외하고 동일
SELECT *
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE product_name = 'Fire-Boltt Ninja Call Pro Plus 1.83" Smart Watch with Bluetooth Call
ing, AI Voice Assistance, 100 Sports Modes IP67 Rating, 240*280 Pixel High Resolution'

# 4개의 product_id 존재, discounted_price에서 1차이 확인, rating 4.3으로 동일
SELECT *
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE product_name = 'Fire-Boltt Phoenix Smart Watch with Bluetooth Calling 1.3",120+ Spor
ts Modes, 240*240 PX High Res with SpO2, Heart Rate Monitoring & IP67 Rating'

# 3개의 product_id 존재
# discounted_price 13,999/ actual_price 19,499/ discount_percentage 0.28/
# rating 4.1/ rating_count 18,998 일치
SELECT *
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE product_name = 'Samsung Galaxy M13 5G (Aqua Green, 6GB, 128GB Storage) | 5000mAh Bat
tery | Upto 12GB RAM with RAM Plus'

# 3개의 product_id 존재
# discounted_price, actual_price, discount_percentage, rating,rating_count 등 차이
# 실질적으로 다른 상품!
SELECT *
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE product_name = 'Firestick Remote'

# 3개의 product_id 존재
# discounted_price 3,999/ actual_price 16,999~17,999/
# discount_percentage 0.76 ~ 0.78/ rating 4.3/ rating_count 17,159 ~ 17,162
SELECT *
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
```

```
WHERE product_name = 'Fire-Boltt Visionary 1.78" AMOLED Bluetooth Calling Smartwatch with
368*448 Pixel Resolution 100+ Sports Mode, TWS Connection, Voice Assistance, SPO2 & Heart
Rate Monitoring'

# 2개의 product_id 존재
# discounted_price 199/ actual_price 399/
# discount_percentage 0.5/ rating 4.2/ rating_count 1,335 - 일치
SELECT *
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE product_name = 'Smashtronics® - Case for Firetv Remote, Fire Stick Remote Cover Cas
e, Silicone Cover for TV Firestick 4K/TV 2nd Gen(3rd Gen) Remote Control - Light Weight/An
ti Slip/Shockproof (Black)'

# 2개의 product_id 존재
# discounted_price 2,199/ actual_price 9,999/
# discount_percentage 0.78/ rating 4.2/ rating_count 29,471 ~ 29,478
SELECT *
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE product_name = "Fire-Boltt India's No 1 Smartwatch Brand Talk 2 Bluetooth Calling Sm
artwatch with Dual Button, Hands On Voice Assistance, 60 Sports Modes, in Built Mic & Spea
ker with IP68 Rating"

# 2개의 product_id 존재
# discounted_price 16,999/ actual_price 24,999/
# discount_percentage 0.32/ rating 4.1/ rating_count 22,318
SELECT *
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE product_name = 'Samsung Galaxy M33 5G (Emerald Brown, 6GB, 128GB Storage) | 6000mAh
Battery | Upto 12GB RAM with RAM Plus | Travel Adapter to be Purchased Separately'

# 2개의 product_id 존재
# discounted_price 12,999/ actual_price 18,999/
# discount_percentage 0.32/ rating 4.1/ rating_count 50,772
SELECT *
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE product_name = 'Redmi Note 11 (Horizon Blue, 6GB RAM, 64GB Storage)|90Hz FHD+ AMOLED
Display | Qualcomm® Snapdragon™ 680-6nm | 33W Charger Included'
```

[결과 이미지] Firestick Remote 의 경우, 가격, 평점 차이 존재 (실질적으로 다른 상품)



※ 위 결과에서 동일한 제품(Firestick Remote를 제외한 나머지 product_name일치)의 경우, 동일데이터로 확인할 수 있었고, 이 경우, 리뷰수를 집계할 때 같은 상품이 중복계산 될 수 있으므로 주의가 필요하다.

- Amazon 플랫폼에서 판매되는 제품은 몇 개의 카테고리로 분류되어 있는가? : 211개의 카테고리 존재

```
SELECT COUNT(DISTINCT category) AS category_count
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
```

[결과 이미지]

행	category_count
1	211

- Amazon 플랫폼에서 판매되는 제품은 몇 개의 카테고리로 분류되어 있는가? : 211개의 카테고리 존재

```
SELECT COUNT(DISTINCT category) AS category_count
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
```

[결과 이미지]

행	category_count
1	211

- 각 카테고리별 제품 수 및 비율 확인
  : 총 211개 카테고리 중 Computers&Accessories|Accessories&Peripherals|Cables&Accessories|Cables|USBCables 카테고리에 161개의 제품이 존재(판매되고 있는 1,351개의 제품(고유값) 중 11.92% 차지)
  ※ 해당 플랫폼에서의 주요 판매 제품 및 카테고리 파악
  - 이커머스에서 어떤 카테고리의 제품이 주요 상품군인지 파악

```
SELECT category,
       COUNT(DISTINCT product_id)AS ProductCountByCategory,
       ROUND(COUNT(DISTINCT product_id)/(SUM(COUNT(DISTINCT product_id)) OVER())*100,2) AS
rate
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
GROUP BY 1
ORDER BY ProductCountByCategory DESC
```

[결과 이미지]

행	category	ProductCountBy...	rate
1	Computers&Accessories|Accessories&Peripherals|Cables&Accessories|Cables|USBCables	161	11.92
2	Electronics|Mobiles&Accessories|Smartphones&BasicMobiles|Smartphones	68	5.03
3	Electronics|WearableTechnolog...	62	4.59
4	Electronics|HomeTheater,TV&Video|Televisions|SmartTelevisions	60	4.44
5	Electronics|Headphones,Earbud...	51	3.77
6	Electronics|HomeTheater,TV&Vi...	49	3.63
7	Home&Kitchen|Kitchen&HomeAppliances|SmallKitchenAppliances|MixerGrinders	27	2.0

- (추가 확인) category에 있어서 '|'으로 세부 카테고리가 나뉘어 있는 것을 확인
  category 를 '|' 기준으로 구분한 후, 대분류(가장 첫번째로 표기된 항목)는 몇개의 항목으로 구분되어 있으며, 분포는 어떻게 되어 있는가?
  : Electronics 526건(35.9%) > Computers&Accessories 453건(30.92%) > Home&Kitchen 448건(30.58%)
  상위 3개의 카테고리에서 전체 판매물품수 1,465건 중 1,427건(총97.4%) 차지

<사용함수 설명>

1. INSTR(str,char) _ 특정 문자열(str)에서 특정 글자('|')가 몇 번째 위치(숫자)를 반환
2. SUBSTR(str,start,len) _ 특정 문자열(str)에서 start 위치부터 len 글자수 추출

```
SELECT
       SUBSTR(category,1, INSTR(category,'|')-1) AS category_lev1,
       COUNT(*) AS product_count,
       ROUND(COUNT(*)*100/(SELECT COUNT(*) FROM `project-5faeab26-b699-4f50-a43.sql_projec
t.recommend`),2) AS Rate
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
GROUP BY 1
ORDER BY product_count DESC;
```

[결과 이미지]

행	category_lev1	product_count	Rate
1	Electronics	526	35.9
2	Computers&Accessories	453	30.92
3	Home&Kitchen	448	30.58
4	OfficeProducts	31	2.12
5	MusicalInstruments	2	0.14
6	HomeImprovement	2	0.14
7	Car&Motorbike	1	0.07
8	Health&PersonalCare	1	0.07
9	Toys&Games	1	0.07

페이지당 결과 수: 50 1 – 9 (전체 9행)

- 전체 판매 상품의 가격 분포 확인
  : discounted_price, actual_price, discount_percentage의 평균값, 최소값, 최대값 확인
  할인가격(discounted_price) 39 ~ 77,990루피 (평균 39루피)
  정상가격(actual_price) 39 ~ 139,900루피 (평균 5,444.99루피)
  할인율(discount_percentage) 0 ~ 94% (평균 47.69%)

```
SELECT
       ROUND(AVG(discounted_price),2) AS disc_price_Avg,
       MIN(discounted_price) AS disc_price_Min,
       MAX(discounted_price) AS disc_price_Max,
       ROUND(AVG(actual_price),2) AS act_price_Avg,
       MIN(actual_price) AS act_price_Min,
       MAX(actual_price) AS act_price_Max,
       ROUND(AVG(discount_percentage)*100,2) AS disc_pct_Avg,
       MIN(discount_percentage)*100 AS disc_pct_Min,
       MAX(discount_percentage)*100 AS disc_pct_Max
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
```

[결과 이미지] 할인율에 있어 평균 47.69%이며 최대 94%의 할인율을 통해 공격적인 할인 정책 확인

행	disc_price_Avg	disc_price_Min	disc_price_Max	act_price_Avg	act_price_Min	act_price_Max	disc_pct_A...	disc_pct_Min	disc_pct_Max
1	3125.31	39.0	77990.0	5444.99	39.0	139900.0	47.69	0.0	94.0

(추가) 전체 판매 상품의 평점(rating) 통계 확인
: rating 컬럼의 데이터 스키마가 STRING으로 이에 대한 FLOAT으로 변환 진행
변환 진행과정에서 오류 문구(Bad double value: |)를 통해 '|'와 같은 문자열이 포함되어 있음 확인
→ 해당 행을 필터링 후 통계 확인 (SAFE_CAST라는 함수를 이용하여, 스키마 변경시 오류 항목 필터링)

※ 함수 설명 SAFE_CAST : CAST 함수 변환시 오류가 발생하는 경우, 멈추는 데 반하여,
SAFE_CAST의 경우, 오류가 발생할 경우 NULL값을 반환하고 실행

```
SELECT
    ROUND(MIN(SAFE_CAST(rating AS FLOAT64)), 2) AS Min_rating,
    ROUND(MAX(SAFE_CAST(rating AS FLOAT64)), 2) AS Max_rating,
    ROUND(AVG(SAFE_CAST(rating AS FLOAT64)), 2) AS Avg_rating,
    COUNT(*)                                     AS total_rows,
    COUNTIF(SAFE_CAST(rating AS FLOAT64) IS NULL) AS excluded_rows
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE rating IS NOT NULL;
```

[결과 이미지] 평점 최소 2.0 ~ 최대 5.0, 평균 4.1 ⇒ 평점이 전반적으로 높음

행	Min_rating	Max_rating	Avg_rating	total_rows	excluded_rows
1	2.0	5.0	4.1	1465	1

(추가) 전체 판매 상품의 리뷰(rating_count) 통계 확인

```
SELECT ROUND(MIN(rating_count),2) AS Min_review,
       ROUND(MAX(rating_count),2) AS Max_review,
       ROUND(AVG(rating_count),2) AS Avg_review
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE rating_count IS NOT NULL
```

[결과 이미지] 리뷰수 최소 2건 ~ 426,973건 (평균 18,296건)

행	Min_review	Max_review	Avg_review
모니터링	2.0	426973.0	18295.54

※ 리뷰수가 많은 상품(product_name 기준) 확인

```
SELECT product_name,category, rating,
       ROUND(MAX(rating_count),2) AS Max_review
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE rating_count IS NOT NULL
GROUP BY 1,2,3
ORDER BY Max_review DESC
LIMIT 10;
```

[결과 이미지]

행	product_name	category	rating	Max_review
1	AmazonBasics Flexible Premium HDMI Cable (Black, 4K@60Hz, 18Gbps), 3-Foot	Electronics|HomeTheater,TV&Video|Accessories|Cables|HDMICables	4.4	426973.0
2	Amazon Basics High-Speed HDMI Cable, 6 Feet - Supports Ethernet, 3D, 4K video,Black	Electronics|HomeTheater,TV&Video|Accessories|Cables|HDMICables	4.4	426973.0
3	Amazon Basics High-Speed HDMI Cable, 6 Feet (2-Pack),Black	Electronics|HomeTheater,TV&Video|Accessories|Cables|HDMICables	4.4	426973.0
4	boAt Bassheads 100 in Ear Wired Earphones with Mic(Taffy Pink)	Electronics|Headphones,Earbuds&Accesso...	4.1	363713.0
5	boAt Bassheads 100 in Ear Wired Earphones with Mic(Furious Red)	Electronics|Headphones,Earbuds&Accesso...	4.1	363713.0
6	boAt Bassheads 100 in-Ear Wired Headphones with Mic (Black)	Electronics|Headphones,Earbuds&Accesso...	4.1	363711.0
7	Redmi 9 Activ (Carbon Black, 4GB RAM, 64GB Storage) | Octa-core Helio G35 | 5000 mAh Battery	Electronics|Mobiles&Accessories|Smartphones&BasicMobiles|Smartphones	4.1	313836.0
8	Redmi 9A Sport (Coral Green, 2GB RAM, 32GB Storage) | 2GHz Octa core Helio G25 Processor | 5000 mAh Battery	Electronics|Mobiles&Accessories|Smartphones&BasicMobiles|Smartphones	4.1	313836.0
9	Redmi 9A Sport (Coral Green, 3GB RAM, 32GB Storage) | 2GHz Octa core Helio G25 Processor | 5000 mAh Battery	Electronics|Mobiles&Accessories|Smartphones&BasicMobiles|Smartphones	4.1	313832.0
10	Redmi 9A Sport (Carbon Black, 2GB RAM, 32GB Storage) | 2GHz Octa core Helio G25 Processor | 5000 mAh Battery	Electronics|Mobiles&Accessories|Smartphones&BasicMobiles|Smartphones	4.1	313832.0

❓"리뷰수가 많고 평점이 높다" = 베스트셀러 상품? :
- 베스트셀러 상품을 확인함으로써 구매자에게 제품을 추천하는데 있어 전략적으로 접근할 수 있다.
- 리뷰수가 많다는 것은 구매 경험자가 많다는 의미이며, 평점은 구매자의 만족도를 나타낸다.
- 이 두 가지 자료를 통하여 '베스트셀러'라고 단정짓기에는 다음과 같은 이유를 들 수 있다.
  1) 실제 판매량에 대한 데이터는 제공되어 있지 않다.
  2) 리뷰수가 편향되었을 수 있다. 리뷰수는 오래 판매된 상품일 수록 리뷰수가 누적되어 있을 수 있다. 실제 판매기간에 대한 리뷰수 비교와 같이 다른 지표가 추가적으로 필요하다.
  3) 현재 주어진 데이터의 경우, 평점 평균이 4.1로 평점만으로 구매자의 만족도 전체를 대표하기에는 부족하다.

- 대분류(category 컬럼에 있어 첫번째로 표기된 항목)별 평균 평점, 할인가격, 할인율 확인

```
SELECT
    SUBSTR(category, 1, INSTR(category,'|') - 1) AS cat_level_1,
    COUNT(*) AS cnt,
    ROUND(AVG(SAFE_CAST(rating AS FLOAT64)), 2) AS avg_rating,
    ROUND(AVG(discounted_price),2) AS avg_disc_price,
    ROUND(AVG(discount_percentage), 1) AS avg_disc_pct
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE rating_count IS NOT NULL
GROUP BY cat_level_1
ORDER BY cnt DESC;
```

[결과 이미지] 상위 4개의 대분류 카테고리에서 결과 도출
(나머지 카테고리의 경우, 제품수가 통계적으로 유의미하지 않음)

- 최고 평균 평점 : OfficeProducts 4.31
- 평균가 최고 : Electronics 5,965.89
- 할인율 최고 : Computers&Accessories 53.9%

행	cat_level_1	cnt	avg_rating	avg_disc_price	avg_disc_pct
1	Electronics	526	4.08	5965.89	50.8
2	Computers&Accessories	451	4.16	845.39	53.9
3	Home&Kitchen	448	4.04	2330.62	40.1
4	OfficeProducts	31	4.31	301.58	12.4
5	MusicalInstruments	2	3.9	638.0	46.0
6	HomeImprovement	2	4.25	337.0	57.5
7	Car&Motorbike	1	3.8	2339.0	42.0
8	Health&PersonalCare	1	4.0	899.0	53.0
9	Toys&Games	1	4.3	150.0	0.0

추천 시스템 구현

1. 가성비 상품 추천

1. 테마 설명 : 고객에게 '가성비템' 추천
   - 고객이 Amazon 사이트에 접속하여 제품을 구경하는 도중 가볍게 구매할 수 있는 제품을 추천하는 시스템으로 특히 평소 살까, 말까 고민했던 상품, 한 번은 사용해보고 싶은 궁금한 제품에 있어서 가격 부담은 없고, 평점이 높은 상품을 추천하는 시스템 구현
2. 구현 로직
   1) 할인율이 50% 이상이며 평점이 4.1 이상이며, 할인된 가격이 620루피(한화 기준 10,000원 이하)이하인 제품 중 상위 제품
   2) '가성비 지표 생성' : 할인율 * 0.6+ 평점 * 0.4
   3) 가성비 지표(value_score) 상위 10개 제품 추

```
SELECT
    product_id,
    product_name,
    SUBSTR(category, 1, INSTR(category,'|') - 1) AS cat_level_1,
```

```
    discounted_price AS sale_price,
    actual_price AS original_price,
    discount_percentage,
    ROUND(SAFE_CAST(rating AS FLOAT64)*20,2) AS rating_num,
    ROUND(discount_percentage * 100 *0.6
        + SAFE_CAST(rating AS FLOAT64) * 20 * 0.4, 2)  AS value_score
FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
WHERE discount_percentage >= 0.50
  AND SAFE_CAST(rating AS FLOAT64) >= 4.1
  AND discounted_price <=620
  AND rating IS NOT NULL
ORDER BY value_score DESC
LIMIT 10;
```

[결과 이미지]

	product_id	product_name	cat_level_1	sale_price	original_price	discount_percent...	rating_num	value_score
1	B097C564GC	rts [2 Pack] Mini USB C Type C Adapter Plug, Type C Female to USB A Male Charger Charging Cable Adapter Converter compatible for iPhone, Samsung S20 ultra/S21/S10/S8/S9/MacBook Pro iPad Silver	Computers&Accessories	294.0	4999.0	0.94	86.0	90.8
2	B0BBW621YC	LAPSTER 12pcs Spiral Cable Protectors for Charger, Wires, Data Charger Cable Protector for Computers, Cell Phones etc (Grey)	Electronics	99.0	999.0	0.9	88.0	89.2
3	B0BP7XLX4B	Syncwire LTG to USB Cable for Fast Charging Compatible with Phone 5/ 5C/ 5S/ 6/ 6S/ 7/8/ X/XR/XS Max/ 11/12/ 13 Series and Pad Air/Mini, Pod & Other Devices (1.1 Meter, White)	Computers&Accessories	399.0	1999.0	0.8	100.0	88.0
4	B08W56G1K9	LAPSTER Spiral Charger Spiral Charger Cable Protectors for Wires Data Cable Saver Charging Cord Protective Cable Cover Set of 3 (12 Pieces)	Computers&Accessories	99.0	999.0	0.9	82.0	86.8
5	B0716ZN31Q	Rts™ High Speed 3D Full HD 1080p Support (10 Meters) HDMI Male to HDMI Male Cable TV Lead 1.4V for All Hdmi Devices Black (10M 30 FEET)	Electronics	598.0	4999.0	0.88	84.0	86.4
6	B0B9LDCX89	STRIFF Mpad Mouse Mat 230X190X3mm Gaming Mouse Pad, Non-Slip Rubber Base, Waterproof Surface, Premium Textured, Compatible with Laser and Optical Mice(Universe Black)	Computers&Accessories	129.0	999.0	0.87	84.0	85.8
7	B0926V9CTV	Elv Mobile Phone Mount Tabletop Holder for Phones and Tablets - Black	Electronics	89.0	599.0	0.85	86.0	85.4
8	B07Q4QV1DL	ELV Aluminum Adjustable Mobile Phone Foldable Tabletop Stand Dock Mount for All Smartphones, Tabs, Kindle, iPad (Black)	Electronics	269.0	1499.0	0.82	90.0	85.2
9	B0BQRJ3C47	REDTECH USB-C to Lightning Cable 3.3FT, [Apple MFi Certified] Lightning to Type-C Fast Charging Cord Compatible with iPhone 14/13/13 pro/Max/12/11/X/XS/XR/8, Supports Power Delivery - White	Computers&Accessories	249.0	999.0	0.75	100.0	85.0
10	B08PPHFXG3	Posh 1.5 Meter High Speed Gold Plated HDMI Male to Female Extension Cable (Black)	Electronics	173.0	999.0	0.83	86.0	84.2

3. 해석
   - 가성비 지표(value_score) 상위 10개의 상품은 최소 75% 의 할인율이 적용된 상품이다.
   - Computers&Accessories와 Electronics 카테고리의 상품이다.
   - product_name을 통해, Mini USB C Type C Adapter Plug, LAPSTER 12pcs Spiral Cable Protectors, STRIFF Mpad Mouse Mat 등과 같이 컴퓨터, 가전 소모품인 것을 확인할 수 있다.
   - 해당 제품의 경우, 쉽게 사용하고 빈번하게 구매하는 소모품으로 평점 4.0 이상 검증된 품질의 제품 중 부담되지 않은 가격의 상품(최종 구매가격 만원 이하)은 목적없이 사이트를 접속했던 고객에게 충분히 매력적인 제안이라고 생각한다.

2. 가격대별 최적 상품 추천

1. 테마 설명 : 고객에게 가격별 선택 기준을 제공한다.
   - 예를 들어, 우리가 친구에게 선물을 할 때, 예산을 정해놓고, 해당 예산 내에서 물건을 구매하듯이, 고객이 예산을 선택하면 해당 예산 범위 내에서 가장 최적(평점이 높고, 리뷰가 많으며, 정가 대비 높은 할인율 반영)의 상품을 추천하는 시스템 구현
2. 구현 로직
   1) 할인가격을 기준으로 고객의 예산을 1단계(600루피, 한화 1만원), 2단계(1800루피, 한화 3만원), 3단계(3000루피, 한화 5만원), 4단계(6000루피, 한화 10만원), 5단계(6000루피 이상, 한화 10만원 이상)으로 구간을 나눈다.
   2) '최고의 선택 지표 생성' : 평점 * 0.5 + 할인율 * 0.4 + 리뷰로그 *0.1
   3) '최고의 선택(best_choice)' 지표 상위 5개 제품 추천

```
WITH price_band AS (
    SELECT
        product_id, product_name,
        SUBSTR(category, 1, INSTR(category,'|') - 1) AS cat_level_1,
        discounted_price AS sale_price,
        actual_price AS original_price,
        discount_percentage,
        ROUND(SAFE_CAST(rating AS FLOAT64)*20,2) AS rating_num,
        rating_count,
        CASE
            WHEN discounted_price <= 600
                THEN '1구간(1만원이하)'
            WHEN discounted_price BETWEEN 601 AND 1800
                THEN '2구간(1만~3만원)'
```

```
            WHEN discounted_price BETWEEN 1801 AND 3000
                THEN '3구간(3만~5만)'
            WHEN discounted_price BETWEEN 3001 AND 6000
                THEN '4구간(5만~10만)'
            ELSE    '고가 (10만원 초과)'
        END AS price_tier,
        ROUND(discount_percentage * 100 *0.5
        + SAFE_CAST(rating AS FLOAT64) * 20 * 0.4 + + LOG(rating_count+1)  * 0.1
        , 2)  AS best_choice
    FROM `project-5faeab26-b699-4f50-a43.sql_project.recommend`
    WHERE rating IS NOT NULL
),
ranked AS (
    SELECT *,
        ROW_NUMBER() OVER (
            PARTITION BY price_tier
            ORDER BY best_choice DESC
        ) AS rnk
    FROM price_band
)
SELECT price_tier, rnk AS rank_in_tier,
       product_id, product_name, cat_level_1,
       sale_price, discount_percentage,
       rating_num, rating_count, best_choice
FROM ranked
WHERE rnk <= 5
ORDER BY price_tier, rnk;
```

[결과 이미지]

: 해당 데이터셋에 구성된 제품이 컴퓨터, 전자 카테고리 제품으로 제품의 다양성은 나타나지 않는다.

1구간 (1만원 이하) : 충전기, 케이블과 같은 소모품으로 구성

행	price_tier	rank_in_tier	product_id	product_name	cat_level_1	sale_price	discount_percent...	rating_num	rating_count	best_choice
1	1구간(1만원이하)	1	B097C564GC	rts [2 Pack] Mini USB C Type C Adapter Plug, Type C Female to USB A Male Charger Charging Cable Adapter Converter compatible for iPhone, Samsung S20 ultra/S21/S10/S8/S9/MacBook Pro iPad Silver	Computers&Accessories	294.0	0.94	86.0	4426	82.24
2	1구간(1만원이하)	2	B0BBWS21YC	LAPSTER 12pcs Spiral Cable Protectors for Charger, Wires, Data Charger Cable Protector for Computers, Cell Phones etc.(Grey)	Electronics	99.0	0.9	88.0	305	80.77
3	1구간(1만원이하)	3	B0BP7XLX48	Syncwire LTG to USB Cable for Fast Charging Compatible with Phone 5/ 5C/ 5S/ 6/ 6S/ 7/8/ X/XR/XS Max/ 11/12/ 13 Series and Pad Air/Mini, Pod & Other Devices (1.1 Meter, White)	Computers&Accessories	399.0	0.8	100.0	5	80.18
4	1구간(1만원이하)	4	B08W56C1K9	LAPSTER Spiral Charger Spiral Charger Cable Protectors for Wires Data Cable Saver Charging Cord Protective Cable Cover Set of 3 (12 Pieces)	Computers&Accessories	99.0	0.9	82.0	8751	78.71
5	1구간(1만원이하)	5	B07182N31Q	Rts™ High Speed 3D Full HD 1080p Support (10 Meters) HDMI Male to HDMI Male Cable TV Lead 1.4V for All Hdmi Devices- Black (10M - 30 FEET)	Electronics	598.0	0.88	84.0	910	78.28

2구간(1만~3만원) : 스마트워치

행	price_tier	rank_in_tier	product_id	product_name	cat_level_1	sale_price	discount_percent...	rating_num	rating_count	best_choice
6	2구간(1만~3만원)	1	B08F54LXW6	Fire Boltt Ninja Call Pro Plus 1.83" Smart Watch with Bluetooth Calling, AI Voice Assistance, 100 Sports Modes IP67 Rating, 240*280 Pixel High Resolution	Electronics	1799.0	0.91	84.0	13937	80.05
7	2구간(1만~3만원)	2	B08F57RN3K	Fire Boltt Ninja Call Pro Plus 1.83" Smart Watch with Bluetooth Calling, AI Voice Assistance, 100 Sports Modes IP67 Rating, 240*280 Pixel High Resolution	Electronics	1799.0	0.91	84.0	13937	80.05
8	2구간(1만~3만원)	3	B08F54972T	Fire Boltt Ninja Call Pro Plus 1.83" Smart Watch with Bluetooth Calling, AI Voice Assistance, 100 Sports Modes IP67 Rating, 240*280 Pixel High Resolution	Electronics	1799.0	0.91	84.0	13937	80.05
9	2구간(1만~3만원)	4	B08F563HB4	Fire Boltt Ninja Call Pro Plus 1.83" Smart Watch with Bluetooth Calling, AI Voice Assistance, 100 Sports Modes IP67 Rating, 240*280 Pixel High Resolution	Electronics	1799.0	0.91	84.0	13937	80.05
10	2구간(1만~3만원)	5	B08F4YBLPX	Fire Boltt Ninja Call Pro Plus 1.83" Smart Watch with Bluetooth Calling, AI Voice Assistance, 100 Sports Modes IP67 Rating, 240*280 Pixel High Resolution	Electronics	1799.0	0.91	84.0	13937	80.05

3구간(3만원~5만원) : 스마트워치

행	price_tier	rank_in_tier	product_id	product_name	cat_level_1	sale_price	discount_percent...	rating_num	rating_count	best_choice
11	3구간(3만~5만)	1	B0B3RRW5F6	Fire-Boltt Phoenix Smart Watch with Bluetooth Calling 1.3",120+ Sports Modes, 240*240 PX High Res with SpO2, Heart Rate Monitoring & IP67 Rating	Electronics	1998.0	0.8	86.0	27709	75.42
12	3구간(3만~5만)	2	B0B3RS90NF	Fire-Boltt Phoenix Smart Watch with Bluetooth Calling 1.3",120+ Sports Modes, 240*240 PX High Res with SpO2, Heart Rate Monitoring & IP67 Rating	Electronics	1999.0	0.8	86.0	27704	75.42
13	3구간(3만~5만)	3	B0B3RSDSZ3	Fire-Boltt Phoenix Smart Watch with Bluetooth Calling 1.3",120+ Sports Modes, 240*240 PX High Res with SpO2, Heart Rate Monitoring & IP67 Rating	Electronics	1999.0	0.8	86.0	27696	75.42
14	3구간(3만~5만)	4	B0B3RRWSF6	Fire-Boltt Phoenix Smart Watch with Bluetooth Calling 1.3",120+ Sports Modes, 240*240 PX High Res with SpO2, Heart Rate Monitoring & IP67 Rating	Electronics	1998.0	0.8	86.0	27696	75.42
15	3구간(3만~5만)	5	B09YV4MW2T	Fire-Boltt India's No 1 Smartwatch Brand Talk 2 Bluetooth Calling Smartwatch with Dual Button, Hands On Voice Assistance, 60 Sports Modes, in Built Mic & Speaker with IP68 Rating	Electronics	2199.0	0.78	84.0	29478	73.63

4구간(5만원 ~ 10만원) : 정수기, 공기청정기, 고급사양의 스마트워치

행	price_tier	rank_in_tier	product_id	product_name	cat_level_1	sale_price	discount_percent...	rating_num	rating_count	best_choice
16	4구간(5만~10만)	1	B08J966M5K	Aquadpure Copper + Mineral RO+UV+UF 10 to 12 Liter RO + UV + TDS ADJUSTER Water Purifier with Copper Charge Technology black & copper Best For Home and Office (Made In India)	Home&Kitchen	4999.0	0.8	92.0	124	77.28
17	4구간(5만~10만)	2	B0B53DS41T	Instant Pot Air Fryer, Vortex 2Q l, Touch Control Panel, 360° EvenCrisp™ Technology, Uses 95 % less Oil, 4-in-1 Appliance: Air Fry, Roast, Bake, Reheat (Vortex 1.9 Litre, Black)	Home&Kitchen	4999.0	0.75	96.0	3964	76.73
18	4구간(5만~10만)	3	B08J6P3LSK	Aqua d pure Active Copper 12-L RO+UV Water Filter Purifier for Home, Kitchen Fully Automatic UF+TDS Controller	Home&Kitchen	4999.0	0.8	90.0	287	76.57
19	4구간(5만~10만)	4	B0B3NDPCS9	Fire-Boltt Visionary 1.78" AMOLED Bluetooth Calling Smartwatch with 368*448 Pixel Resolution 100+ Sports Mode, TWS Connection, Voice Assistance, SPO2 & Heart Rate Monitoring	Electronics	3999.0	0.78	86.0	17161	74.38
20	4구간(5만~10만)	5	B0B3N7LR6K	Fire-Boltt Visionary 1.78" AMOLED Bluetooth Calling Smartwatch with 368*448 Pixel Resolution 100+ Sports Mode, TWS Connection, Voice Assistance, SPO2 & Heart Rate Monitoring	Electronics	3999.0	0.76	86.0	17159	73.38

5구간(10만원 초과) : 공기청정기, 외장하드, TV

행	price_tier	rank_in_tier	product_id	product_name	cat_level_1	sale_price	discount_percent...	rating_num	rating_count	best_choice
20	4구간(5만~10만)	5	B0B3N7LR6K	Fire-Boltt Visionary 1.78" AMOLED Bluetooth Calling Smartwatch with 368*448 Pixel Resolution 100+ Sports Mode, TWS Connection, Voice Assistance, SPO2 & Heart Rate Monitoring	Electronics	3999.0	0.76	86.0	17159	73.38
21	고가 (10만원 초...	1	B08GJ57MKL	Coway Professional Air Purifier for Home, Longest Filter Life 8500 Hrs, Green True HEPA Filter, Traps 99.99% Virus & PM 0.1 Particles, Warranty 7 Years (AirMega 150 (AP-1019C))	Home&Kitchen	14400.0	0.76	88.0	3837	74.03
22	고가 (10만원 초...	2	B08CTYFC37	SanDisk 1TB Extreme Portable SSD 1050MB/s R, 1000MB/s W,Upto 2 Meter Drop Protection with IP55 Water/dust Resistance, HW Encryption, PC,MAC & TypeC Smartphone Compatible, 5Y Warranty, External SSD	Computers&Accessories	10989.0	0.68	88.0	41398	70.26
23	고가 (10만원 초...	3	B09X1M9DHX	iFFALCON 80 cm (32 inches) HD Ready Smart LED TV 32I 53 (Black)	Electronics	9999.0	0.64	84.0	1269	66.31
24	고가 (10만원 초...	4	B09ZPM4C2C	TCL 80 cm (32 inches) HD Ready Certified Android Smart LED TV 32S5205 (Black)	Electronics	10901.0	0.65	82.0	398	65.9
25	고가 (10만원 초...	5	B09ZPJT8B2	TCL 80 cm (32 inches) HD Ready Certified Android Smart LED TV 32S615 (Black)	Electronics	11990.0	0.63	84.0	64	65.52

3. 해석
   - 해당 데이터셋의 97.4% 이상 Computers&Accessories, Electronics, Home&Kitchen 카테고리에 분포되어 있어, 제공되는 제품이 다양하지 않은 것을 확인할 수 있다.
   - 1구간(만원이하)에서는 가볍게 구매할 수 있는 소형 전자 소모품, 2~3구간에서는 스마트워치, 4구간에서는 정수기, 공기청정기, 고사양 스마트워치가 추천되고 있으며, 5구간(고가)에서는 TV,외장하드, 공기청정기가 최적 상품으로 추천된다.
   - 고객의 예산에 맞추어 제품의 할인율, 평점, 리뷰수를 활용하여 구매 당사자의 선호가 아닌 받는 사람의 필요를 예측한 구매가 가능할 것으로 예상된다.